

Sample Question Paper: Python Programming (CSE2025)

End Semester Examination

Academic Year: 2024-25, Batch: 2024, 2nd Semester

Duration: 2 Hours Maximum Marks: 40

Instructions

- Answer all questions.
- Each question carries equal marks.
- Write clear and concise code.
- Ensure that your code is well-indented and readable.

Questions

1. **Input/Output and Data Types** Write a Python program that takes a user's name and age as input and prints a message saying "Hello [name], you are [age] years old." Ensure that the age is handled as an integer.
2. **Operators** Write a Python program that takes two integers as input and prints "Yes" if the first integer is divisible by the second and greater than the second, otherwise prints "No".
3. **Data Structures** Write a Python program that prompts the user to enter a list of words separated by commas. Store these words in a list, remove any duplicates by converting the list to a set, and then create a dictionary where each word is a key and its length is the value. Finally, print the dictionary.
4. **Control Flow** Write a Python program that uses a loop to print the first 10 numbers in the Fibonacci sequence.
5. **Functions** Define a Python function named `is_palindrome` that takes a string as input and returns `True` if the string is a palindrome and `False` otherwise.
6. **Modules (Random)** Write a Python program that imports the `random` module and uses it to generate and print a random integer between 1 and 100, inclusive.
7. **File Handling** Write a Python program that reads from a text file named `input.txt` and writes all lines that contain the word `python` (case-sensitive) to another file named `output.txt`. Assume that `input.txt` exists in the same directory as your program.
8. **Data Analysis (NumPy)** Using the `NumPy` library, create an array containing the numbers from 0 to 9, inclusive. Then, reshape this array into a 2x5 matrix and print it.

9. **Exception Handling** Write a Python program that prompts the user to enter two numbers. The program should then divide the first number by the second and print the result. Use a try-except block to handle the case where the second number is zero, and print an appropriate error message.
10. **Modules (Date & Time)** Write a Python program that prompts the user to enter two dates in the format YYYY-MM-DD and calculates the number of days between them. Use the datetime module.
11. **Data Analysis (Pandas)** Write a Python program that uses Pandas to create a DataFrame from a dictionary containing student names and their marks (e.g., {"Name": ["Alice", "Bob", "Charlie"], "Marks": [85, 90, 78]}). Calculate and print the average marks.
12. **Data Visualization (Seaborn)** Using Seaborn and a given list of data (e.g., [10, 20, 25, 30, 35]), write a Python program to create a boxplot and save it as boxplot.png. Ensure the plot has a title "Data Distribution".
13. **Modules (Math)** Write a Python program that imports the math module and calculates the area of a circle given its radius as user input. Print the result rounded to two decimal places.